

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
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MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK PROTOCOLS & ARCHITECTURE LAB
EXPERIMENT # 07

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Score: _____ Signature of the Lab Tutor: _____ Date: _____

Implementing Point-to-Point Protocol on Serial Interfaces

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives:

As a result of this lab section, you should achieve the following tasks:

- Establish PPP encapsulation as a serial link layer protocol
- Set the link layer protocol to PPP on interface Serial 1/0/0 of R1 and R2.
- Assign an IP address to interface Serial 1/0/0 of R1 and R2.

Lab Overview:

In this lab, we aim to establish PPP (Point-to-Point Protocol) as the encapsulation method on a serial connection between two routers. By default, most routers use HDLC encapsulation on their serial interfaces, but PPP is more widely used because it is vendor-neutral and supports additional features such as authentication and error detection. In this exercise, we will configure the serial interface Serial 1/0/0 on both R1 and R2 to use PPP instead of the default protocol. Once the encapsulation method is set, we will assign IP addresses to the serial interfaces so that both routers can communicate with each other at the network layer. Finally, connectivity will be tested to ensure that the PPP link has been established successfully. This lab will help you understand the role of PPP as a data link layer protocol, how to configure it on real routers, and how it enables reliable communication across WAN serial links.

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Point-to-point Protocol—PPP:

The Point-to-Point Protocol (PPP) is a data link layer (Layer 2) communication protocol that is used to establish a direct connection between two networking devices, usually routers, over serial links. PPP provides a standard method for encapsulating network layer protocols such as IP, IPv6, and others so that they can be transmitted across a serial connection.

Unlike HDLC, which is often vendor-specific, PPP is an open standard protocol, meaning it can be used between devices from different manufacturers, making it more versatile and widely adopted in WAN (Wide Area Network) environments.

Key Features:

1. Encapsulation:

PPP encapsulates network layer packets (like IP datagrams) inside its own frame format. Each PPP frame contains fields for addressing control, protocol identification, data (payload), and error detection.

2. Error Detection:

PPP uses Cyclic Redundancy Check (CRC) to detect errors in transmitted frames. This ensures that corrupted data can be identified and discarded, providing reliable communication.

3. Authentication:

PPP supports authentication methods such as PAP (Password Authentication Protocol) and CHAP (Challenge Handshake Authentication Protocol). These mechanisms allow routers to verify the identity of the device on the other end of the link before establishing communication, which adds a layer of security.

4. Multi-Protocol Support:

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PPP can carry multiple types of network layer protocols simultaneously through its Protocol field in the frame.

5. Flexibility in WAN Links:

PPP is commonly used in leased lines, dial-up connections, DSL connections, and other WAN technologies. It allows ISPs and organizations to connect their customers or branch offices reliably over serial or point-to-point connections.

PAP (Password Authentication Protocol):

PAP is a simple authentication protocol used in PPP connections. It works by sending the username and password in plain text from the client to the server. Since the credentials are not encrypted, PAP is considered less secure and can be vulnerable to eavesdropping. However, it is easy to implement and was one of the earliest methods of authentication in PPP.

CHAP (Challenge Handshake Authentication Protocol)

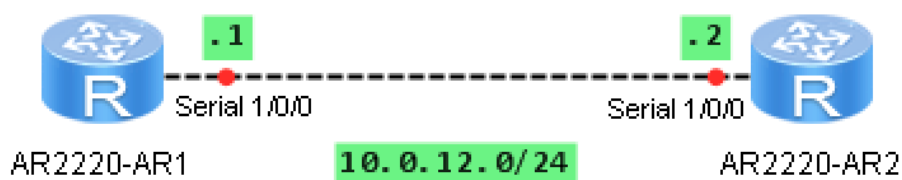
CHAP is a more secure authentication protocol used in PPP connections. Instead of sending the password directly, it uses a challenge-response mechanism. The server sends a random challenge to the client, the client responds with a value created using the password and a hash function (like MD5), and the server verifies it. This prevents the password from being sent over the link. CHAP also periodically re-authenticates during the session, making it more secure than PAP.

Serial Communication:

Serial communication is the process of sending data one bit at a time, sequentially, over a communication channel. In WANs, serial interfaces are used to connect routers across long distances. PPP operates over these serial links to provide reliable point-to-point communication.

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Topology



Scenario

Routers R1 and R2 are connected through a PPP link. The network administrator wants to set the IP address of Serial 1/0/0 on R1 to 10.0.12.1/24, and Serial 1/0/0 on R2 to 10.0.12.2/24 so that IP packets can be transmitted through the PPP link.

Tasks

Step 1 Configure Router 1

```
<Huawei>system-view
```

Enter system view, return user view with Ctrl+Z.

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```
[Huawei]sysname R1
[R1] interface serial 1/0/0
[R1-Serial1/0/0] link-protocol ppp
[R1-Serial1/0/0] ip address 10.0.12.1 24
```

Step 2 Configure Router 2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2] interface serial 1/0/0
[R2-Serial1/0/0] link-protocol ppp
[R2-Serial1/0/0] ip address 10.0.12.2 24
```

Step 3 Verify the configuration

```
[R2-Serial1/0/0]ping 10.0.12.1
  PING 10.0.12.1: 56 data bytes, press CTRL_C to break
    Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=170 ms
    Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=60 ms
Reply from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=30 ms      Reply
from 10.0.12.1: bytes=56 Sequence=4 ttl=255 time=50 ms          Reply from
10.0.12.1: bytes=56 Sequence=5 ttl=255 time=60 ms

  --- 10.0.12.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
    round-trip min/avg/max = 30/74/170 ms
```

Configuration Roadmap

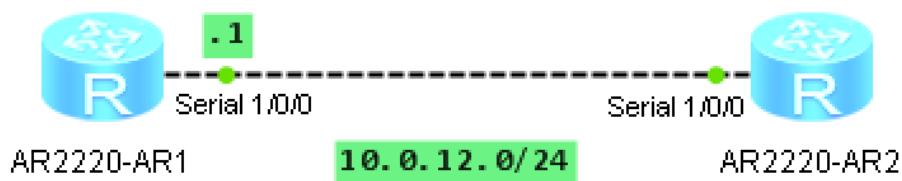
The configuration roadmap is as follows:

1. Set the link layer protocol to PPP on interface Serial 1/0/0 of R1 and R2.
2. Set the IP address of Serial 1/0/0 on R1 to 10.0.12.1, and configure R1 to assign IP address 10.0.12.2 to R2.

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3. Configure R2 to obtain an IP address from R1 for Serial 1/0/0.

Topology



Scenario

The network administrator now wants to set the IP address of Serial 1/0/0 on R1 to 10.0.12.1/24 and allow R1 to assign IP address 10.0.12.2 to Serial 1/0/0 on R2, enabling the link between R1 and R2 to transmit IP packets.

Tasks

Step 1 Configure Router 1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1] interface serial 1/0/0
[R1-Serial1/0/0] link-protocol ppp
[R1-Serial1/0/0] ip address 10.0.12.1 24
[R1-Serial1/0/0] remote address 10.0.12.2
```

Step 2 Configure Router 2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2] interface serial 1/0/0
```

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```
[R2-Serial1/0/0] link-protocol ppp
[R2-Serial1/0/0] ip address ppp-negotiate
```

Step 3 Verify the configuration

```
[R2]display interface serial 1/0/0
Serial1/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2019-06-06 06:08:44 UTC-08:00
Description: HUAWEI, AR Series, Serial1/0/0 Interface
Route Port, The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is negotiated, 10.0.12.2/32
Link layer protocol is PPP
LCP opened, IPCP opened
Last physical up time : 2019-06-06 06:05:26 UTC-08:00
Last physical down time : 2019-06-06 06:05:18 UTC-08:00
Current system time: 2019-06-06 06:10:09-08:00
Physical layer is synchronous, Virtualbaudrate is 64000 bps
Interface is DTE, Cable type is V11, Clock mode is TC
Last 300 seconds input rate 7 bytes/sec 56 bits/sec 0 packets/sec
Last 300 seconds output rate 2 bytes/sec 16 bits/sec 0 packets/sec

Input: 69 packets, 2242 bytes
  Broadcast:          0, Multicast:          0
Errors:              0, Runts:              0
Giants:              0, CRC:                0
  Alignments:         0, Overruns:          0
Dribbles:            0, Aborts:             0   No
Buffers:             0, Frame Error:        0

Output: 69 packets, 830 bytes
  Total Error:        0, Overruns:          0
  Collisions:         0, Deferred:          0
  Input bandwidth utilization :    0%
Output bandwidth utilization :    0%

[R2]ping 10.0.12.1
  PING 10.0.12.1: 56 data bytes, press CTRL_C to break
    Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=200 ms
```

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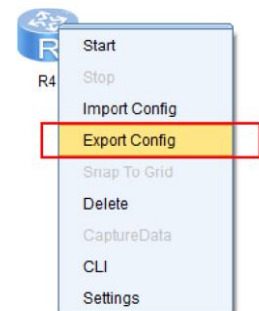
```
Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=40 ms
Reply from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=30 ms
Reply from 10.0.12.1: bytes=56 Sequence=4 ttl=255 time=50 ms
Reply from 10.0.12.1: bytes=56 Sequence=5 ttl=255 time=50 ms
```

```
--- 10.0.12.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 30/74/200 ms
```

Step 4 Save the topology and configuration

Upload the .topo .efz .cfg files

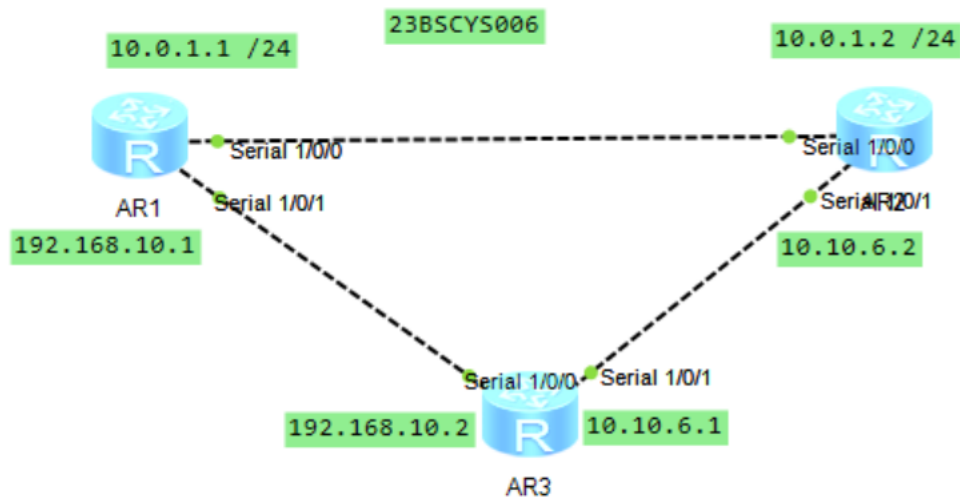
- ☐ run the **save** command to save the current configuration on each device
<Huawei>save
- ☐ **Export** the devices' configuration
- ☐ click the **Save/Save as** button on the Tools bar to save the whole project.



Tasks

Use three routers and perform PPP link.

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```
AR1
AR1 AR2 AR3
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
 firewall zone Local
  priority 15
#
 interface Serial1/0/0
  link-protocol ppp
  remote address 10.0.12.2
  ip address 10.0.1.1 255.255.255.0
#
 interface Serial1/0/1
  link-protocol ppp
  remote address 192.168.10.2
  ip address 192.168.10.1 255.255.255.0
#
 interface GigabitEthernet0/0/0
#
 interface GigabitEthernet0/0/1
#
 interface GigabitEthernet0/0/2
#
 interface NULL0
#
 ip route-static 10.10.6.0 255.255.255.0 192.168.10.2
 ip route-static 192.168.10.0 255.255.255.0 192.168.10.2
#
 user-interface con 0
  authentication-mode password
 user-interface vty 0 4
```

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```
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw)%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface Serial1/0/0
link-protocol ppp
ip address ppp-negotiate
#
interface Serial1/0/1
link-protocol ppp
ip address ppp-negotiate
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
ip route-static 192.168.10.0 255.255.255.0 10.0.1.1
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
```

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```
AR3
AR1  AR2  AR3
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
firewall zone Local
 priority 15
#
interface Serial1/0/0
 link-protocol ppp
 ip address ppp-negotiate
#
interface Serial1/0/1
 link-protocol ppp
 remote address 10.10.6.2
 ip address 10.10.6.1 255.255.255.0
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
ip route-static 10.0.1.0 255.255.255.0 10.10.6.2
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
```